



Lateral root formation and nutrients: nitrogen in the spotlight

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Abstract

Lateral roots are important to forage for nutrients due to their ability to increase the uptake area of a root system. Hence, it comes as no surprise that lateral root formation is affected by nutrients or nutrient starvation, and as such contributes to the root system plasticity. Understanding the molecular mechanisms regulating root adaptation dynamics toward nutrient availability is useful to optimize plant nutrient use efficiency. There is at present a profound, though still evolving, knowledge on lateral root pathways. Here, we aimed to review the intersection with nutrient signaling pathways to give an update on the regulation of lateral root development by nutrients, with a particular focus on nitrogen. Remarkably, it is for most nutrients not clear how lateral root formation is controlled. Only for nitrogen, one of the most dominant nutrients in the control of lateral root formation, the crosstalk with multiple key signals determining lateral root development is clearly shown. In this update, we first present a general overview of the current knowledge of how nutrients affect lateral root formation, followed by a deeper discussion on how nitrogen signaling pathways act on different lateral root-mediating mechanisms for which multiple recent studies yield insights.

Nutrients shape the root system, but how?

Nutrients seem to be important in the control of lateral root formation as the availability of mineral nutrients or the lack of them affect lateral root formation, at least in certain conditions and depending on the species and/or growth stage (Gruber et al., 2013; Motte et al., 2019). A major problem in interpreting the influence of nutrients on lateral root formation is the fact that increasing levels of nutrients will affect the overall fitness of the plants including the root system, whereas both nutrient deficiency and excess may cause severe stress to the plants. Hence, it is not always clear whether the observed effects on the root system are caused

directly by a sensing event or indirectly by a nutritional effect. The use of biotechnological tools, such as mutant lines specifically defective in nutrient signaling, could help deciphering the true nature of the nutrient effect but requires a thorough insight into the respective pathways. At least for phosphate and nitrogen, two major nutrients, such knowledge is currently available and argues for a clear and direct effect on lateral root formation (see below). But the involvement of other nutrient signaling pathways cannot be unambiguously demonstrated.

For instance for potassium (K^+), another major nutrient (Zorb et al., 2014), our current knowledge is scarce. K^+ starvation has in most cases an inhibitory effect on first-order

ADVANCES

- All nutrients affect lateral root development, but only for nitrogen important pathways are clearly unraveled.
- NRT1.1 is a multi-role nitrate transceptor: a nitrate transporter with two different affinities, a nitrate receptor, and an auxin transporter, and all these functions impact nitrate-mediated lateral root development.
- Nitrate-mediated tissue-specific post-transcriptional regulation and phosphorylation status affecting its sub-cellular localization brings another level of complexity on the role of NRT1.1 in lateral root development.
- Nitrate-controlled auxin transport and biosynthesis regulate lateral root primordium development and emergence.
- Ammonium acts on lateral root development by modulating the apoplastic pH, modifying the auxin mobility within cells, and easing or impairing lateral root emergence.

lateral roots, but a stimulatory effect on second-order lateral roots (Kellermeier et al., 2014; Sustr et al., 2019). Here, at least the ARABIDOPSIS K⁺ TRANSPORTER1 (AKT1) seems to be essential for K⁺ starvation-induced second-order lateral root and primary root growth responses (Kellermeier et al., 2014; Li et al., 2017). This occurs independently of K⁺ accumulation, suggesting a role of AKT1 as a K⁺-sensor. It is, moreover, shown that K⁺ starvation results in degradation of the auxin transporter PIN-FORMED1 (PIN1) in an AKT1-dependent manner (Li et al., 2017). The hormone auxin is dominantly involved in lateral root formation (see below). Hence, the K⁺ effect on auxin transport might explain the effect of K⁺ starvation on lateral root formation as well but awaits further research. Sulfur deficiency inhibits lateral root formation and involves CLAVATA1 (CLV1), a peptide receptor involved in lateral root development (Araya et al., 2014; Dong et al., 2019), but it is not known how this occurs. Iron, an essential micronutrient for which presumptive sensing and signaling components are partially reported (Kobayashi, 2019), exerts in high concentrations a clear inhibitory effect on lateral root initiation, possibly by reducing PIN2 expression (Li et al., 2015). A link with the iron signaling network is not clear, so this is not per se caused by iron sensing. Iron also interacts with phosphate (Müller et al., 2015; Balzergue et al., 2017; Dong et al., 2017; Mora-Macías et al., 2017; Godon et al., 2019) and affects phosphate starvation-mediated lateral root development (Rai et al., 2015), which might partially explain its effect on lateral root formation. Aluminum interacts in a similar way with phosphate (Godon et al., 2019): its stimulatory effect on lateral root formation seems to be mediated

through phosphate starvation signaling (Ruíz-Herrera and López-Bucio, 2013). Phosphate starvation results in higher auxin levels and expression of the auxin receptor TRANSPORT INHIBITOR RESPONSE1 (TIR1) in lateral root primordia (Nacry et al., 2005; Pérez-Torres et al., 2008), stimulating their development. The central transcription factor PHOSPHATE STARVATION RESPONSE1 (PHR1), which is activated specifically upon phosphate deficiency, might play a central role here. A comparison between PHR1-activated genes and lateral root formation genes revealed multiple candidates that might be directly targeted and contribute to the induction of lateral roots upon phosphate starvation, including TIR1, and the auxin signaling AUXIN RESPONSE FACTOR7 (ARF7) and ARF19 (Crombez et al., 2019). However, their importance still has to be experimentally validated. In summary, the involvement of several nutrients in lateral root formation awaits further investigation. Deeper insight has been obtained for nitrogen and more specifically nitrate and ammonium. In contrast to other nutrients, several studies unraveled how nitrogen signaling affects different lateral root developmental stages. For this reason, in the following we limit ourselves to the discussion on pathways that are modulated by nitrogen signaling affecting lateral root formation.

Nitrogen affects multiple processes during lateral root formation

Nitrate levels can vary widely in different spots only millimeters apart in the soil (Lark et al., 2004). Nitrate is a highly mobile nutrient and tends to leach with water runoffs, while ammonium has a lower leaching tendency as it binds to the cation exchange complexes present in organic matter and clay particles in soils due to its positive charge. Next to mobility, other factors such as uneven microbial activity (Nunan et al., 2002) or organic matter distribution and soil structure (van Noordwijk et al., 1993) affect nitrogen distribution in soils. A remarkable adaptation to nutrient heterogeneity is the foraging behavior that favors root system development in nutrient-rich patches rather than in nutrient-poor patches conferring a strong developmental advantage in situations of interplant competition. Foraging behavior is regulated by local and systemic nutrients-derived signals and hormones in a complex way (Zhang et al., 1999; Ruffel et al., 2011; Guan et al., 2014; Tabata et al., 2014; Ristova et al., 2016; Guan et al., 2017; Ohkubo et al., 2017; Poitout et al., 2018). As such, plants adopted different strategies in response to the differential nitrate and ammonium distribution and are able to adapt their root system to this changing environment during their growth cycle.

Root plasticity is an asset for plants to cope with uneven nitrogen distribution and to compete with other plants for nutrient-rich patches. Understanding the molecular mechanisms regulating root adaptation to heterogeneous nutrient distribution is a prerequisite to develop root systems with better nutrient use efficiency. The lateral root branching pattern is a key element of the plasticity of root systems in

response to nitrogen. This branching pattern, on its turn, strongly relies on the formation and outgrowth of lateral roots from the parent root. The level, distribution and form of nitrogen affect different lateral root developmental stages: nitrate stimulates not only lateral root initiation, but also primordium development and lateral root emergence, low nitrate levels still promote primordium development, but too low or locally low levels actively inhibit development and emergence, whereas too high levels block lateral root emergence. Finally, local ammonium levels stimulate lateral root emergence. Here, we review the current knowledge on how the two main nitrogen forms, nitrate and ammonium, affect these different stages of lateral root formation.

Lateral root initiation

In *Arabidopsis* (*Arabidopsis thaliana*), the entire process of lateral root development runs through multiple sequential steps and mainly involves auxin signaling (Lavenus et al., 2013; Du and Scheres, 2018; Santos Teixeira and Ten Tusscher, 2019). At the very start, a pre-patterning process specifies sites in the primary root where lateral roots can be produced (Xuan et al., 2020). Here, lateral roots originate from pericycle cells, which are primed to become lateral root founder cells by an auxin signaling mechanism (De Smet et al., 2007; Dubrovsky et al., 2008; De Rybel et al., 2010). Persisting auxin signaling results in a first asymmetric division of the founder cells followed by a series of well-orchestrated divisions leading to the formation of a lateral root primordium (De Smet et al., 2007; Goh et al., 2016; von Wangenheim et al., 2016).

Nitrate induces auxin signaling in the pericycle

Nitrate treatment positively affects lateral root initiation, which occurs by modulating auxin signaling. More specifically, nitrate induces in the pericycle area the expression of the auxin receptor *AUXIN SIGNALING F-BOX3* (*AFB3*), which is required for nitrate-dependent lateral root initiation (Vidal et al., 2010). *AFB3* induction leads to the start of auxin signaling reflected by the degradation of the *SOLITARY ROOT/INDOLE-3-ACETIC ACID INDUCIBLE14* (*SLR/IAA14*) transcriptional repressor. This results in the induction of the *NAC DOMAIN-CONTAINING PROTEIN4* (*NAC4*), encoding a transcription factor essential for the nitrate-induced lateral root initiation (Vidal et al., 2013). *SLR/IAA14* is a known master regulator of lateral root initiation (Fukaki et al., 2002; Vanneste et al., 2005). It interacts with *ARF7* and *ARF19*, which can trigger the first asymmetric division of the founder cells leading to the formation of a lateral root primordium (Fukaki et al., 2005; Okushima et al., 2007; Kang et al., 2013). Surprisingly, *NAC4* expression is still induced by nitrate in *arf7/19* single and double mutants, suggesting that other ARFs may contribute as well to this pathway downstream of *SLR/IAA14* (Vidal et al., 2013). Despite its clear importance for nitrate-induced lateral root initiation, it is still unknown how nitrate induces *AFB3* expression. The *NITRATE TRANSPORTER1.1* (*NRT1.1*)

transceptor, which has functions in nitrate sensing, nitrate transport and auxin transport (Liu et al., 1999; Liu and Tsay, 2003; Krouk et al., 2010; Parker and Newstead, 2014; Sun et al., 2014; Bouguyon et al., 2015; Wang et al., 2020; see Box 1), has a key role in most nitrate responses. However, in contrast to most nitrate-responsive genes, the induction of *AFB3* or *NAC4* is independent of the *NRT1.1* signaling function. Nevertheless, *NRT1.1* is important for this pathway, but only as a nitrate transporter (Liu et al., 1999; Vidal et al., 2014), possibly to direct nitrate to the pericycle cells. Finally, nitrate metabolites induce the expression of the microRNA *miR393* that negatively controls *AFB3* transcript levels (Vidal et al., 2010). Hence, the *AFB3/miR393/NAC4* module integrates external nitrate signals, internal nitrogen status and auxin signaling to regulate lateral root initiation in the root in response to nitrate (Figure 1A).

The nitrate-induced lateral root initiation seems to be dependent on the nitrate-induced transcription factors *TGACG SEQUENCE-SPECIFIC BINDING PROTEIN1* (*TGA1*) and *TGA4* as well. Like the *AFB3/miR393/NAC4* module, *TGA1/4* gene induction by nitrate requires *NRT1.1*-mediated nitrate transport (Alvarez et al., 2014), but in contrast to *AFB3*, *TGA1/4* are both regulated via the *NRT1.1* Ca^{2+} -signaling pathway (see Box 2; Marchive et al., 2013; Riveras et al., 2015; Alvarez et al., 2020; Wang et al., 2020). The *TGA* transcription factors directly induce expression of the nitrate transporters *NRT2.1* and *NRT2.2*, which are also required for the nitrate-induced lateral root initiation (Little et al., 2005; Remans et al., 2006; Alvarez et al., 2014). Besides, *NRT2.1* is also positively regulated in nitrate-rich patches due to systemic signaling involving C-terminally encoded peptide (CEP) family members or cytokinin transport, but seems to mainly affect nitrate uptake and root elongation in this context (Ruffel et al., 2011; Tabata et al., 2014; Ohkubo et al., 2017; Poitout et al., 2018; Figure 1A).

Finally, the *ARF8/miR167* module is also involved in nitrate-mediated lateral root initiation in the root pericycle. Nitrate, after reduction into glutamine, inhibits the level of *miR167*. *miR167* is an *ARF8* repressor, thus its inhibition results in higher levels of *ARF8* which, in turn, regulate lateral root initiation (Gifford et al., 2008), which a pathway that seems to be conserved in other species (Wang et al., 2015; Figure 1A).

Lateral root primordia development

Following the first asymmetric cell division of lateral root founder cells, a series of successive periclinal and anticlinal divisions lead to the formation of a dome-shaped lateral root primordium (Goh et al., 2016; von Wangenheim et al., 2016). The lateral root progression depends on auxin transport and a gradual change in the expression and polar localization of PIN auxin efflux carriers. This eventually results in a root meristem with an inversed auxin fountain flow, where shoot derived and locally produced auxin radiates from the base to the tip of the lateral root primordium causing an apical auxin maximum, before being redistributed toward

Box 1 NRT1.1 IS A MULTI-FACETED NITRATE TRANSECTOR

NRT1.1 (also referred to as CHL1 or NPF6.3) is a major actor in nitrate-related responses which, at least in *Arabidopsis*, but most likely in other plants as well (Wang et al., 2020), serves different functions:

- (1) A nitrate receptor. Nitrate binds NRT1.1 to induce a nitrate-dependent signaling pathway. It triggers Ca^{2+} signaling and activates different Ca^{2+} -sensor protein kinases (CPKs). These CPKs phosphorylate the NIN-LIKE PROTEIN6 (NLP6) and NLP7 transcription factors and prevent their nuclear export, inducing their nuclear accumulation where they trigger the transcription of hundreds of nitrate-responsive genes (Marchive et al., 2013; Guan et al., 2017; Liu et al., 2017; Alvarez et al., 2020). Notably, *NRT1.1* itself is induced by NLP7 (Marchive et al., 2013), increasing both its nitrate sensing and other roles (see below), but obstructing the study of specific subfunctions of NRT1.1. Moreover, the NRT1.1 phosphorylation status on its Thr101 residue triggers a switch between low- and high-affinity conformations in response to external nitrate concentrations with implications on NRT1.1 transport, sensing and signaling functions (see below). The phosphorylation state also affects the subcellular trafficking of NRT1.1, and as such its auxin transport function and probably its signaling function as well (see below; Zhang et al., 2019).
- (2) A dual-affinity nitrate transporter. NRT1.1 is a nitrate transporter and hence, in addition to its signaling function, may be involved in nitrate-dependent responses by influencing nitrate distribution and local levels. Just as for the sensing function, the phosphorylation state of the NRT1.1 Thr101 residue dictates its transport activity, switching from a High-Affinity Transport System (HATS) when external nitrate concentration is in the micromolar range to a Low-Affinity Transport System (LATS) when concentration increases to the millimolar range (Liu et al., 1999; Liu and Tsay, 2003; Parker and Newstead, 2014; Sun et al., 2014).
- (3) An auxin transporter. Besides nitrate, NRT1.1 transports auxin. As this only occurs with the Thr101 phosphorylated NRT1.1 form under low nitrate conditions, the auxin transport function is at the end regulated by nitrate (Krouk et al., 2010; Bouguyon et al., 2015). The NRT1.1-dependent auxin transport is facilitated due to the recruitment of the Thr101 phosphorylated NRT1.1 form into the plasma membrane (Zhang et al., 2019).

the base via the outer cell layers (Benková et al., 2003; Marhavý et al., 2014; Du and Scheres, 2017; see Box 1).

Nitrate starvation blocks primordia development

Under local or systemic nitrate-poor conditions, lateral root primordia development is impaired and results in prematurely arrested primordia (Krouk et al., 2010; Mounier et al., 2014), preventing the roots from developing into nitrogen-poor soil regions. The auxin transport function of NRT1.1 (see Box 2) is important in blocking this development: NRT1.1 increases the auxin reflux from the lateral root primordium tip, preventing the apical auxin accumulation and, hence, lateral root emergence. With increasing external nitrate concentrations, NRT1.1 auxin transport activity becomes inhibited, thus restoring the auxin accumulation at the primordium tip and stimulating its progression (Krouk et al., 2010; Bouguyon et al., 2015). However, the nitrate-induced expression of *NRT1.1*, which is occurring in lateral root primordia as well (Mounier et al., 2014; Bouguyon et al., 2016) is paradoxical in this model. Apparently, a tissue-specific nitrate-dependent posttranscriptional regulatory mechanism dictates the final abundance of the NRT1.1 protein. As a result, NRT1.1 protein levels are, specifically from Stage III primordia onwards, repressed by nitrate, even though *NRT1.1* expression is upregulated while it is not the case in nitrate-poor conditions (Bouguyon et al., 2016; Figure 1B).

Moreover, nitrate-dependent phosphorylation of NRT1.1 affects its subcellular localization on the plasma membrane

and the subsequent NRT1.1-mediated auxin dynamics (Zhang et al., 2019). Plants with nonphosphorylatable mutant versions of *NRT1.1* are defective in lateral root NRT1.1-mediated basipetal auxin transport (reflux of auxin from the lateral root tip) normally observed upon nitrogen deprivation, and hence accumulate auxin in the lateral root primordium. On the other hand, constitutive phosphorylation of the Thr101 residue led to constitutive basipetal auxin transport and inhibits lateral root development. The phosphorylation state may affect the electrostatic and mobility properties of NRT1.1 in the cell, affecting its spatiotemporal dynamics and ability to transport auxin. The phosphorylated form of NRT1.1 accumulates at the plasma membrane microdomain and facilitates auxin transport while the non-phosphorylated form of NRT1.1 has increased mobility which eases its internalization via the clathrin-mediated pathway. This concurs with the observed reduced NRT1.1 occurrence upon high nitrate in lateral root primordia (Bouguyon et al., 2016).

The growth arrest of lateral root primordia due to increased basipetal auxin transport is further reinforced by the suppression of auxin biosynthesis in the stele under limiting nitrogen conditions. More specifically, this is due to nitrate starvation and NRT1.1-dependent negative transcriptional regulation of *TRYPTOPHAN AMINOTRANSFERASE-RELATED PROTEIN2* (*TAR2*), encoding an auxin biosynthesis enzyme (Maghiaoui et al., 2020). Nitrate on the other hand nullifies the NRT1.1-dependent repression of *TAR2* in particular in

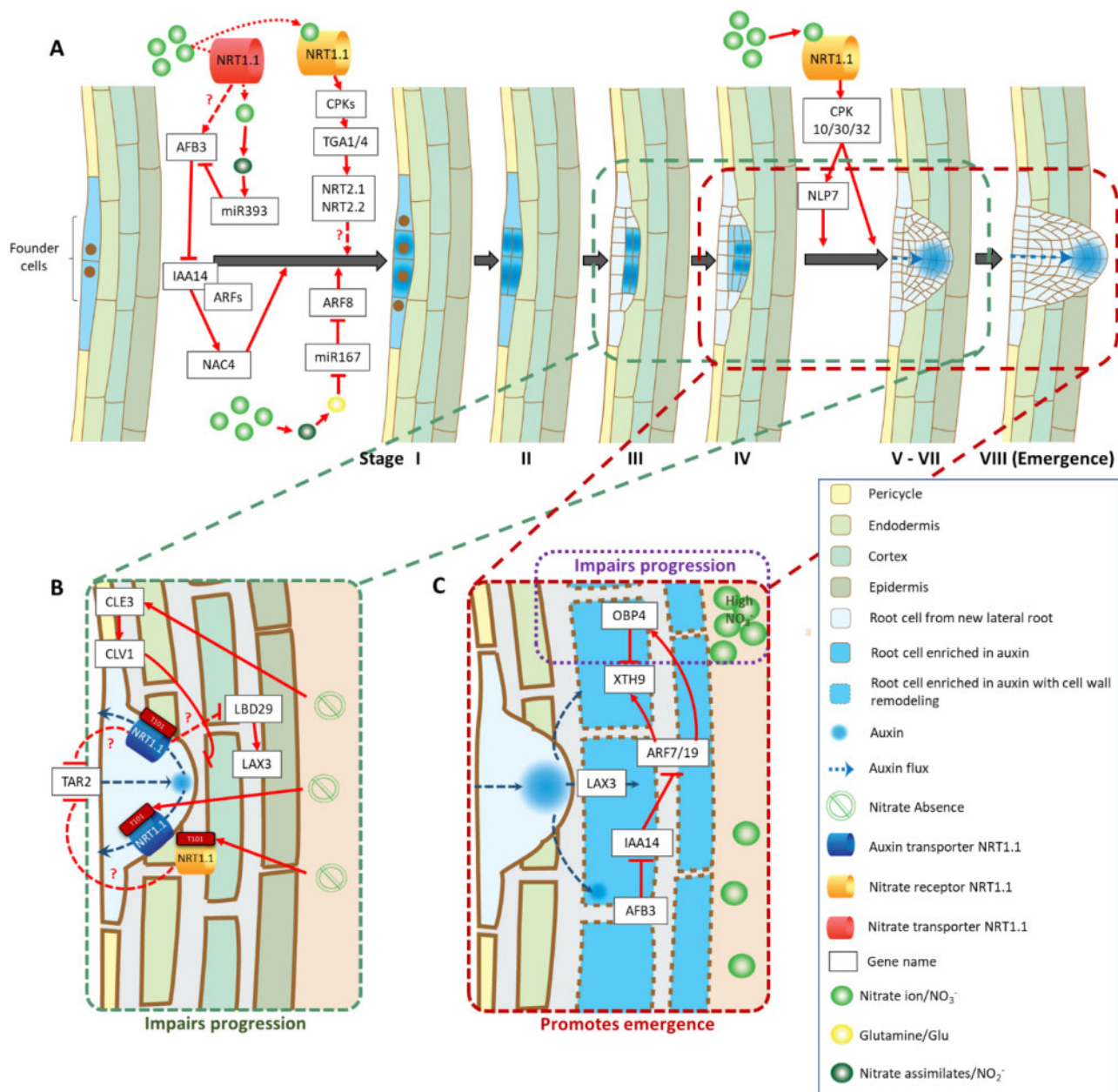


Figure 1 Nitrate affects lateral root initiation and the development of the lateral root primordium. A, The lateral root primordium passes through different developmental stages (Roman numerals according to Malamy and Benfey, 1997) from its initiation to emergence. Regulatory mechanisms, mostly mediated by NRT1.1, regulate different steps during lateral root formation depending on the availability and levels of the nutrient. See details in the text for the depicted pathways. B, Details on how nitrate deprivation impacts auxin distribution in the developing lateral root primordium. Under nitrate deprivation, NRT1.1 is, due to post-transcriptional regulation, highly present specifically in the lateral root primordium. Here, its auxin transport function refluxes auxin to the base of the developing primordium impairing its progression. In parallel, the nitrate receptor NRT1.1 represses the auxin biosynthesis enzyme TAR2 in the stele impairing auxin accumulation and lateral root primordium development while it also represses the auxin transporter LAX3 preventing auxin distribution to the overlying cells and lateral root emergence (see panel C). Finally, in the pericycle, CLE3 is induced and blocks, via its receptor CLAVATA1, the progression of the primordium. C, Upon nitrate availability, NRT1.1 is not present anymore in the lateral root primordium and hence does not refrain auxin accumulation and diffusion from the tip to the overlying cells. The auxin transporter LAX3 is not blocked which leads to an increase of auxin levels in the surrounding cells resulting in the induction of genes such as the AFB3-dependent XTH9 involved in the modification of cell walls to facilitate the emergence of the primordium from the parent root. In the case nitrate levels are very high; however, OBP4 inhibits XTH9 expression downstream of the AFB3 pathway, impairing the primordium outgrowth.

the root stele and young lateral root primordia (Stages I–III), but not in later stages from Stage IV onwards. In contrast, NRT1.1, or at least its auxin transport function, is known to act from Stage III onwards under low nitrate, hence at later

stages than the NRT1.1-dependent repression of TAR2. Thus, upon nitrate deficiency, NRT1.1 prevents auxin accumulation in lateral root primordia by either increasing the basipetal auxin transport and/or by repressing TAR2

Box 2 AUXIN IN LATERAL ROOT MERISTEM ESTABLISHMENT AND EMERGENCE

Auxin, when sensed by TRANSPORT INHIBITOR RESPONSE1/AUXIN SIGNALING

F-BOX (TIR1/AFB) auxin receptors, promotes the degradation of Aux/IAA repressors which in turn frees auxin response factors that induce auxin responsive genes. Different IAA/ARFs combinations are possible and are referred to as ‘modules’ (Lavenus et al., 2013; Du and Scheres, 2018). Regarding lateral root development, one module is predominant: SLR/IAA14-ARF7-ARF19 being involved in all steps of lateral root formation, but other modules intervene as well (reviewed in Santos Teixeira and Ten Tusscher (2019)).

The SLR/IAA14-ARF7/19 auxin module regulates, possibly via the LATERAL ORGAN BOUNDARIES DOMAIN16 (LBD16), LBD18, and LBD29 transcription factors, a number of PLETHORA (PLT) genes (Feng et al., 2012; Du and Scheres, 2017), which are important for the formative divisions within the lateral root primordium, for correct expression of root meristem genes and meristem establishment and for the expression pattern of PIN genes encoding auxin efflux transporters (Du and Scheres, 2017). The latter is important to achieve an auxin maximum in the young primordia and, after re-localization of the PINs, a maximum in the primordium tip. This auxin maximum is, together with PLT expression, important for meristem maintenance and correct development and outgrowth of the primordium (Benková et al., 2003; Marhavý et al., 2014; Du and Scheres, 2017).

Interestingly, the dome-shape results from mechanical constraints from the overlying cells rather than via an organized division pattern (Lucas et al., 2013). Next to the PLT genes, the expression of PUCHI, a gene involved in the lateral root primordium boundaries definition, is also regulated downstream of the SLR/IAA14-ARF7/19 (Kang et al., 2013).

Around Stage IV, the endodermis adapts its volume and shape to ease the primordium lateral root progression (Vermeer et al., 2014). In the cortical and epidermal cells overlaying the endodermis, auxin activates the AUXIN TRANSPORTER- LIKE PROTEIN3 (LAX3) downstream of the SLR/IAA14-ARF7/19 module and the LBD29 transcription factor. LAX3 imports auxin from the primordium, which accumulates in the overlying cells in a positive feedback loop, inducing cell wall remodeling enzymes and changing the volume and turgor of the surrounding cells by modulating aquaporins (PIPs), which together mechanically ease the primordium emergence through the primary root cells layers (Swarup et al., 2008; Peret et al., 2012; Vermeer et al., 2014; Porco et al., 2016; Banda et al., 2019).

expression, depending on the lateral root developmental stage (Figure 1B).

Another pathway that inhibits lateral root primordia development in nitrate-deficient conditions depends on CLAVATA3/ESR-RELATED (CLE) peptides CLE1, -3, -4, and -7 and their receptor CLAVATA1 (CLV1). Nitrogen starvation induces the expression of these CLE peptides, which are putting a stop to lateral root primordia development at the early Stages I–IV (Araya et al., 2014). In *Medicago* (*Medicago truncatula*), different CLE-peptides inhibit nodulation (a process with strong resemblance to lateral root development), acting downstream of a nitrate-induced NLP transcription factor (Nishida et al., 2018). Likewise, still in *Medicago*, CEPs are involved in the negative regulation of lateral root development by nitrate (Imin et al., 2013). In *Arabidopsis*, CEPs are induced upon local nitrogen starvation and are an important signal at the basis of the systemic nitrate signaling (Tabata et al., 2014), while they also negatively regulate lateral root development (Roberts et al., 2016). Hence, CEPs might be important to prevent lateral roots in nitrogen-poor regions, but this link has not explicitly been shown yet.

The NRT1.1-CPK-NLP cascade is involved in nitrate-induced lateral root development

NRT1.1 is, as a nitrate receptor, known to be the entry point of a nitrate signaling cascade mediated by CALCIUM-DEPENDENT PROTEIN KINASE10 (CPK10), CPK30, and

CPK32, and their phosphorylation of the NLP6 and 7 transcription factors (see Box 2; Riveras et al., 2015; Liu et al., 2017). This pathway seems important for the induction of lateral roots formation upon nitrate treatment, as mutation of the three CPK genes leads to a reduction of nitrate-induced lateral root density. In contrast to the CPK-independent AFB3 pathway affecting initiation and emergence, the NRT1.1-CPK-NLP signaling cascade primarily affects primordia development: the triple *cpk10/30/32* mutant shows a higher density of lateral root primordia at the developmental Stages II–IV, but a lower number of emerged lateral root primordia and developed lateral roots (Liu et al., 2017). Likewise, NLP7-overexpression lines have an increased lateral root density (Yu et al., 2016). At present, it is not clear how the NRT1.1-CPK-NLP pathway affects lateral root primordia development. NLP7 binds directly to the promoters of several genes involved in lateral root primordia development, and much more seems to be indirectly regulated (Marchive et al., 2013; Alvarez et al., 2020). A notable example of a gene directly targeted by NLP7 is PIN7 (Alvarez et al., 2020). PIN7 encodes an auxin transporter and is important for lateral root formation. Remarkably, just as the *cpk* triple mutant, *pin7* mutants show a reduction in lateral roots but with an increase in Stages III and IV primordia (Marhavý et al., 2013). Another possible important player downstream of NLP7 and a possible explanation for the observed *cpk* phenotype is TAR2 (Alvarez et al., 2020), important for auxin

biosynthesis in the root stele during nitrate-induced lateral root formation and upregulated by nitrate in an *NRT1.1*-dependent manner (see also above; Maghiaoui et al., 2020). Recently, it was shown that nitrate dephosphorylate the PIN2 auxin efflux carrier in a *NRT1.1*-dependent manner (Ötvös et al., 2021). Although PIN2 is in particular associated with outgrowth of the lateral root (Benkova et al., 2003), the phosphorylated form is downregulated during lateral root morphogenesis (Zhang et al., 2013), and possibly imply yet another control of lateral root formation by nitrate.

MADS-box transcription factors impact nitrate-dependent lateral root development

Different MADS-box transcription factors are important in the nitrate-regulated development of lateral root primordia. In rice (*Oryza sativa*), the MADS-box TRANSCRIPTION FACTORS25, 27a/b, and 57 (*OsMADS25*, 27a/b, and 57) expression levels are induced by nitrate (Puig et al., 2013) and their transcript accumulation is repressed by the monocot-specific miRNA *miR444a* or its isoform *miR444a.2* (Li et al., 2010; Jiao et al., 2020). In presence of nitrate, *OsMADS25* inhibits the expression of *OsIAA14*, an Aux/IAA transcriptional repressor of ARFs, which promotes auxin accumulation in the lateral root primordium to induce its progression (Zhang et al., 2018). Several other genes involved in auxin biosynthesis and polar transport are differentially regulated under low and high nitrate in *OsMADS25* mutants and over-expression lines, indicating that *OsMADS25* targets a broad range of genes involved in lateral root development (Zhang et al., 2018). *OsMADS27* increases lateral root density in response to nitrate via ABA signaling (Chen et al., 2018), which is known to affect lateral root primordia progression most likely via influencing auxin response, biosynthesis, and transport (De Smet et al., 2003; Antoni et al., 2013; Shkolnik-Inbar et al., 2013; Xing et al., 2016). Similarly, the AGAMOUS-LIKE21 (*AGL21*) MADS-box transcription factor in Arabidopsis, belonging to the same clade as *OsMADS25*, 27b, and 57, increases auxin biosynthesis in the stele and young primordia to stimulate lateral root formation in low nitrate conditions (Yu et al., 2014).

Lateral root emergence

A growing lateral root primordium has to grow through the overlying cell layers in order to emerge from the main root, whereas the resistance of those overlying cells is an important factor to control lateral root primordia progression (Peret et al., 2012; Lucas et al., 2013; Vermeer et al., 2014; Banda et al., 2019). To allow lateral root emergence, reprogramming of the overlying cells in response to auxin is essential (see Box 2). Auxin-related mechanisms, including auxin accumulation in the lateral root primordium and auxin transport from primordium toward the overlying cells are required to enable lateral root emergence (Marchant et al., 2002; Benková et al., 2003; Swarup et al., 2008; Marhavý et al., 2014).

Nitrate affects different processes to stimulate lateral root emergence

A crucial transporter importing auxin to the cortical and epidermal cells overlying lateral root primordia is the auxin transporter LAX3 (Swarup et al., 2008), which is induced by nitrate but downregulated in absence of nitrate specifically in lateral root primordia and downstream of *NRT1.1* (Maghiaoui et al., 2020). *NRT1.1* is specifically absent in lateral root primordia upon nitrate treatment (see also above; Bouguignon et al., 2016), and is hence only suppressing LAX3 in nitrate-deficient conditions. As a result, there is no auxin accumulation in the overlying cells, and emergence of lateral roots is blocked if the nitrogen level is too low, avoiding waste of energy by producing superfluous lateral roots. *NRT1.1*-dependent signaling seems to repress the expression of *LBD29*, a direct positive regulator of LAX3 involved in lateral root development (Porco et al., 2016; Maghiaoui et al., 2020; see Box 1). The signaling cascade downstream of *NRT1.1* leading to the inhibition of *LBD29* is still unclear. It is possible that the auxin reflux mediated by *NRT1.1* under low nitrogen in the lateral root primordium tip lowers auxin diffusion in the surrounding cells, impairing *LBD29* and LAX3 expression (Figure 1, B and C).

One of the enzymes controlling lateral root emergence by loosening cell walls of the cells surrounding the lateral root primordium is the XYLOGLUCAN ENDOTRANSGLUCOSYLASE/HYDROLASE9 (*XTH9*). *XTH9* is induced during lateral root emergence, and *xth9* mutants show a decreased lateral root density but an increased number of pre-emerged lateral root primordia stages. *XTH9* expression is induced by nitrate downstream of the nitrate-induced AFB3 auxin receptor (Xu and Cai, 2019), which is independent of *NRT1.1*-signaling (Vidal et al., 2010, 2014). Downstream of AFB3, *XTH9* expression is dependent on the auxin signaling ARF7 and ARF19 transcription factors (Xu and Cai, 2019). This illustrates how the same lateral root pathway can be affected by nitrate in different ways: under nitrate-deficient conditions, the presence of *NRT1.1* in lateral root primordia inhibits the auxin-inducible expression of *LBD29* via its auxin reflux mechanism, whereas at a higher nitrate level, *LBD29* expression is most likely stimulated due to the nitrate-dependent induction of AFB3 (see above). At very high nitrate concentrations however, a feedback regulation occurs depending on the AFB3/SLR/IAA14-ARF7/19 signaling module where the transcription factor OBF BINDING PROTEIN4 (*OBP4*) negatively regulates *XTH9* expression by binding directly to its promoter (Xu and Cai, 2019). As a result, very high nitrate levels impair lateral root emergence (Figure 1C). Moreover, other genes involved in cell wall remodeling such as *EXPANSIN17* (*EXP17*), important for lateral root emergence as well (Lee and Kim, 2013), were found differentially regulated in *obp4* mutants and *OBP4*-overexpressing lines (Xu et al., 2016). These genes may play a role as important as *XTH9* during the nitrate mediated lateral root formation.

In rice, NITRATE ASSIMILATION RELATED2.1 (*OsNAR2.1*), which is induced by nitrate downstream of *NRT1.1* and known to interact with the *NRT2.1* nitrate transporter to

mediate high-affinity nitrate uptake, also interacts with and is required for the induction of the *NITRILASE1* (*OsNIT1*) and (*OsNIT2*) expression in response to nitrate (Song et al., 2020). Single and double mutants analysis of *osnar2.1*, *osnit1*, and *osnit2* showed that these genes have an overlapping function in increasing lateral root density upon nitrate supply. *OsNIT1* can convert indole-3-acetonitrile (IAN) into indole-3-acetic acid (IAA), a crucial step in auxin biosynthesis, and this enzymatic activity is further increased by interaction with *OsNIT2* and *OsNAR2.1*. Both *osnit* and *osnar2.1* mutants also show a decreased shoot-to-root auxin transport, which could explain the reduction of lateral root density observed in these mutants (Huang et al., 2015; Song et al., 2020). As shoot-derived auxin is particularly important for lateral root emergence (Bhalerao et al., 2002; Swarup et al., 2008), the *NAR2.1*-dependent pathway looks to be an additional way for nitrate to control lateral root emergence. This control seems to be used by relatively low nitrate concentrations and is conserved in Arabidopsis: both in Arabidopsis and rice, *nar2.1* mutants show a reduction of lateral root primordia density specifically under low nitrate conditions (Orsel et al., 2006; Huang et al., 2015).

Remarkably, *OsNIT1* and *OsNIT2* are transcriptionally induced by ammonium (Song et al., 2020) unlike *OsNAR2.1* that is downregulated upon ammonium supplementation (Feng et al., 2011; Song et al., 2020). Under low ammonium (0.25 mM), *osnit1* and *osnit2* mutants display a 35%–45% lower lateral root density compared to the wild type and a reduction in shoot-to-roots transport of IAA. Shoot-derived auxin was recently shown to be important for ammonium-induced lateral root emergence (see below; Meier et al., 2020) and hence this mechanism may be shared between ammonium and nitrate-controlled lateral root emergence.

Ammonium affects lateral root emergence by changing the apoplastic pH and auxin movement toward the overlying cells

Ammonium, another important nitrogen source for crop production, also affects lateral root development. Local ammonium patches in particular increase higher order lateral root branching (Lima et al., 2010). But contrary to nitrate for which several gene regulatory pathways regulating root architecture have been unraveled and described in the past decades, a mechanism for ammonium-induced lateral root formation was only recently discovered. Using an auxin response reporter, Meier et al. (2020) found that upon local ammonium nutrition a shoot-derived increased auxin signal in cell layers overlying lateral root primordia at Stages II–VI is taking place, both known to be important for lateral root emergence (see above; Bhalerao et al., 2002; Swarup et al., 2008). The auxin signal depends on ammonium transport as it is not observed in the quadruple knock-out *qko* mutant, lacking the four most important ammonium transporters (Meier et al., 2020). The increase of auxin in the overlying cells preceding lateral root emergence is associated with LAX3 and AUXIN RESISTANT1 (*AUX1*)-dependent auxin

transport (Marchant et al., 2002; Swarup et al., 2008), but ammonium-mediated auxin distribution in cells overlying the lateral root primordium appears to bypass these pathways as the *aux1lax3* mutant presents a higher density of lateral roots in presence of ammonium (Meier et al., 2020). This indicates that an alternative mechanism increases the auxin transport toward the overlying cells. As ammonium uptake releases protons in the apoplast via H^+ -ATPases (Figure 2), it leads to the protonation of IAA into IAAH. This form of auxin is able to diffuse passively through cell-membranes (Delbarre et al., 1996; Esteban et al., 2016), leading Meier et al. (2020) to assume an important contribution of the acidification of the apoplast upon ammonium treatment. Indeed, buffering the growth medium to an acidic pH (5.0) induces an ammonium-like lateral root phenotype, both in wild-type plants as in the ammonium transport defective *qko* mutant, whereas an alkaline pH in ammonium-fed plants prevents lateral root density increase. Altogether, these results validate the ammonium uptake-dependent mechanism that leads to apoplast acidification and modifies auxin mobility, resulting in an increased auxin accumulation in the cells overlying the primordium and activating the programs to enable the lateral root emergence (Meier et al., 2020; Figure 2). Hence, in contrast to other regulatory mechanisms, classically having a signal/receptor mechanism to induce a biochemical pathway, ammonium is rather “sensed” due to its effect on intercellular pH and primarily induces a chemical pathway (Motte and Beeckman, 2020; see Outstanding Questions). Interestingly, a similar mechanism was recently discovered in rice. Contrary to Arabidopsis, rice is an ammonium tolerant plant growing in hypoxic paddy fields rich in ammonium. In a similar fashion as in Arabidopsis, ammonium uptake induce H^+ -ATPase-mediated acidification of rice root tissues which induces root phenotypes that again can be mimicked by buffering the growth medium into acidic pH, and which depends on auxin distribution as well (Jia et al., 2020), showing the presence of similar pH-dependent mechanisms between the two species. Although Jia et al. (2020) did not focus on lateral roots, the acidification seems to increase the number of lateral roots as well, arguing for a conserved mechanism of ammonium induced lateral root formation. Besides, multiple NRTs are H^+ -coupled nitrate symporters, which function in pH homeostasis (Fang et al., 2016; Feng et al., 2020) and which explains the, as opposed to ammonium, alkalization of the apoplast after nitrate treatment (Meier et al., 2020).

Conclusion and perspectives

Lateral root development follows a series of complex mechanisms, from the priming of lateral root founder cells in the basal meristem of the primary root to the emergence of a new lateral root. As sessile organisms, plants evolved a highly plastic root system to adapt to nutrient heterogeneity in the soil. Similar to the growing insights into nitrogen signaling, knowledge on the pathways that control root branching is continuously growing (Motte et al., 2019).

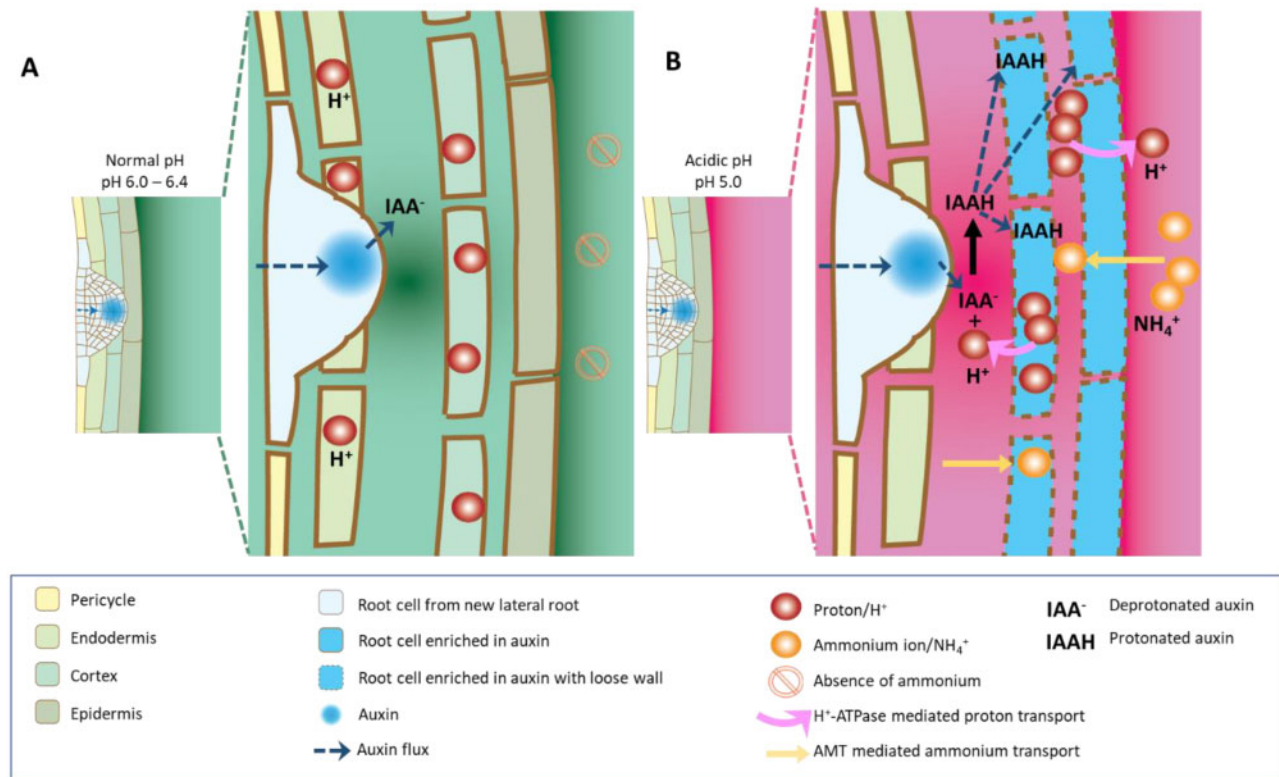


Figure 2 Ammonium induces pH changes in the root apoplast and the rhizosphere, which in turn modifies the auxin protonated state affecting its mobility. A, In absence of ammonium, the apoplast pH is at its normal physiological level. The lateral root primordium tip derived auxin is less protonated due to low levels of protons in the apoplast, and this form of auxin diffuses less to the surrounding cells. B, Ammonium increases the H⁺ content of the apoplast, hence decreasing its pH, which leads to auxin protonation. Protonated auxin diffuses easily to the overlying cells surrounding the developing lateral root primordium, triggering cell wall remodeling enzymes and lateral root emergence.

Multiple studies unraveled important cross-talks between nitrate, and recently ammonium, with lateral root development. Nitrate especially triggers, notably via its NRT1.1 transceptor, signaling cascades at each step of the lateral root development. These cascades are induced by nitrate itself, by the uptake of nitrate or by modulating NRT1.1 auxin transport function. It functions not only by modifying the lateral root initiation in the pericycle cells, but also by promoting or inhibiting the lateral root primordium growth and by modulating the plasticity of cells surrounding the developing lateral root. Compared to nitrate, less is known about ammonium but recent advances brought a step forward in understanding how this ion regulates lateral root development. While nitrate induces molecular and hormonal pathways, ammonium induces chemical changes within the root and in particular the apoplast, where acidification modulate auxin signals and eases or refrains lateral root emergence. Hence, one single nutrient induces several signals affecting lateral root development, which depends on the level or distribution of nitrogen.

Whereas the nitrogen effect on lateral root development mainly concern post-founder cell specification effects, recent findings yielded important insights into the processes affecting and preceding founder cell specification (Xuan et al., 2020). The effect of nutrients on these mechanisms is not

known yet but might unravel more cross-talking pathways. Still, studying the effects of external cues on early events is often technically challenging and may partially explain why our knowledge on the cross-talk between nutrients other than nitrogen and lateral root development is scarce whereas the effects are clear (Gruber et al., 2013; Motte et al., 2019).

Another level of complexity comes from the overlap of different nutrient signaling pathways, which is best documented for nitrate and phosphate that interact at multiple levels (recently reviewed by Krouk and Kiba (2020)). Nitrate for example activates via NRT1.1 the SPX DOMAIN-CONTAINING PROTEIN4 (SPX4)/PHR1-mediated phosphate starvation signaling cascade and controls the phosphate starvation response (Hu et al., 2019; Medici et al., 2019), whereas phosphate starvation modulates nitrate uptake (Ueda et al., 2020; Wang et al., 2020). As such, the phosphate starvation-induced lateral root formation might even follow a nitrogen path. Finally, recent evidences showed that NRT1.1 is important for K⁺ uptake and allocation from the root to the shoots (Fang et al., 2020). The importance of NRT1.1 in the distribution of a nutrient that also affects lateral root formation (see above) highlights once more the complex interaction and overlaps between different nutrients and their signaling pathways, and on their

OUTSTANDING QUESTIONS

- To what extent does cross-talk between various nutrient signaling pathways impact lateral root development?
- Which signal dictates the presence or absence of the NRT1.1 protein in the lateral root primordium at specific developmental stages?
- How do ammonium and AMTs control shoot-derived auxin transport in the lateral root primordium?
- Does ammonium only impact the root system via an indirect chemical signaling effect or are there other signaling pathways in play?

probable interaction to regulate root development. The highly connected nitrate, potassium and phosphate responses open perspectives on the deciphering of mechanisms by which nutrients co-interact to modulate root growth, and such insights will prove to be useful for applications in breeding plants that are more efficient in nutrient uptake in farming situations where fertilization often implies a cocktail of nutrients, or to develop ideotypes adapted to particular situations. In the example of phosphate, plants foraging in the top-soil where phosphate is concentrated perform better than plant developing a large root system (Lynch, 2011), whereas in the case of nitrogen, a deep root foraging seeking to reach leaching nitrate (known as the “steep, cheap and deep” ideotype) may perform better than top soil foraging under certain circumstances (Lynch, 2013). Improving crop yields by considering the root system in the breeding programs is currently underexploited but very promising in the short future (Den Herder et al., 2010; Lynch, 2019). Focusing on only one nutrient has so far yielded mitigated results on improving plant fitness (Garnett et al., 2009), and we expect that an integrated approach of nutrients interaction will catalyze the transition from laboratories to fields. With the current technical advances in imaging methods and plant biotechnological tools, we anticipate a growing interest in the field of root-nutrient cross-talks and look forward to new discoveries filling the knowledge gaps in plant nutritional research (see Outstanding Questions).

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